



Exploring view-specific label relationships for multi-view multi-label feature selection

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ABSTRACT

In the domain of multi-view multi-label (MVML) learning, features are distributed across various views, each offering multiple semantic representations. While existing approaches aim to balance commonality and complementarity within the view space, the inconsistency in label space has been underexplored, revealing the inadequacy of assuming uniform labels. Thus, there is a pressing need to explore the relationship among view-specific labels. Our method diverges from previous approaches by focusing on imposing constraints tailored to learning view-specific labels. We aim to preserve common characteristics through inter-view relationships while retaining specific traits inherent to each view's instances. By employing these strategies, we establish a clear mapping between labels and feature representations, enabling precise feature weighting. Convergence to the optimal feature set is achieved through multiplicative updating rules. Our method demonstrates superiority across most of cases through comprehensive experimental analysis compared to existing state-of-the-art alternatives.

1. Introduction

As information retrieval becomes more convenient, the representation of data exhibits a diversity of forms. Notably, in computer vision, objects can be observed from multiple viewpoints or angles, each offering a distinct “view” with complementary features and characteristics [1,2]. Likewise, in natural language processing, language can be scrutinized from different vantage points, including syntax, semantics, and discourse [3]. The fusion of these multiple views through multi-view learning can enhance the comprehensiveness of information. Unfortunately, the dimension increases with more views in MVML learning, rendering the exploration of discriminative features a more arduous task [4,5].

Currently, most existing methods in MVML learning focus on extracting the common representation across all views. Some representative works involve extracting the common part from the feature matrix in each view [6,7]. Another works involve considering the label matrix itself as the common part to solve the common problem in view space [8]. Nonetheless, these existing methods often neglect to exploit the complementarity offered by each view. Consequently, the specific features present in each view are often

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Table 1
The distinction between representative works and the proposed method.

Method	Common	Complementary	View-specific labels	Lable noises	Label relationship
IrMMC [6]	✓				
M2LD [7]	✓				
ELSMML [8]	✓			✓	
LSA-MML [15]	✓	✓			
ICM2L [11]	✓	✓			
MVLD [12]	✓	✓	✓		
I^2 VSLC	✓	✓	✓	✓	✓

ignored, potentially leading to suboptimal results in the mapping process. This, in turn, can impact the feature weights and the result in deviations in the feature selection process.

As a result, recent researches have been dedicated to incorporating complementary information from multiple views. The specific part not only contributes to the common part, but also emphasizes its own distinctive features. Although there are researches in the field of unsupervised multi-view feature selection that balance both aspects [9,10], the utilization of label information poses challenges due to the inherent unknown of labels. However, in MVML learning, it is noteworthy that these methods assume an ideal uniform label matrix, thus limiting their practical applicability [11,12]. A comparative analysis of the disparities between previous and our novel method is concluded in Table 1.

Furthermore, the issue of non-alignment garners more attention, particularly within the realm of multiple views. Accessing aligned information can prove challenging in practice [13]. Moreover, the assumption is inappropriate that all views have a uniform set of labels [14]. Consequently, view-specific labels are defined, wherein each view has its corresponding label matrix, and each view-specific label contains a portion of the observed label matrix. Given the specific characteristics of the MVML scenario, it is crucial to consider the view-specific label, which is convenient to deal with more scenarios.

To overcome these challenges, the innovative method called Intra-View and Inter-View of View-Specific Label Correlation for Multi-View Multi-Label Feature Selection (I^2 VSLC) is proposed. Initially, we establish mappings between the features of each view and their corresponding view-specific labels. Subsequently, the learning process for view-specific labels incorporates both commonalities and complementarities, striving to achieve similarity between views while simultaneously preserving their distinctiveness. It is noteworthy that the ground-truth matrix for the labels can be decomposed into view-specific label matrices, which is a prerequisite for learning the view-specific label. Furthermore, the feature matrix is constrained at different hierarchies to guide the view-selection and the feature-selection. This hierarchical constraint ensures the effectiveness and robustness of the feature selection process. Extensive experiments demonstrate the superiority for I^2 VSLC. Compared with existing methods, the method has the following main contributions:

- The method I^2 VSLC employs a distinct mapping for each view and establishes the relationship between view-specific label matrices from different hierarchies, encompassing intra-view and inter-view correlations.
- The proposed method delves into the correlation between view-specific labels and capitalizes on the equivalence between view-specific labels and ground-truth labels. This strategy aims to strike a balance between the distinctiveness and shared characteristics of the labels, with the ultimate goal of selecting informative features that simultaneously consider the commonality and complementarity with view-specific labels.
- Conducting comprehensive experiments on six datasets to evaluate the efficiency of our method I^2 VSLC, and its exceptional capabilities is substantiated through analyzing the final results.

The paper remained is organized as follows. Section 2 provides an overview of the related research. Subsequently, in Section 3, comprehensively describes the implementation procedure of the proposed method I^2 VSLC. Experimental results are analyzed in Section 4. Finally, the conclusion is given in Section 5.

2. Related work

Generally, MVML learning can be viewed as an extension of multi-label learning. Effectively harnessing the information embedded in label correlations is crucial for achieving optimal performance in multi-label learning [16]. Thus, the related work introduces label correlation learning first, and then gives a brief for the work about MVML learning.

2.1. Label correlation learning

In the realm of label relevance computation, it can be categorized into three distinct groups. The first category falls under first-order methods like Binary Relevance (BR) [17] and Pruned Problem Transformation (PPT) [18], which assume label independence and employ transformation-based techniques to tackle multi-label challenges. However, it's worth noting that this type of methods often leads to the creation of numerous new classes. Another notable method, ML-kNN, the actual theory of which is based on the k -nearest neighbor algorithm and does not necessitate the introduction of novel classes. ML-kNN leverages statistical information

and adheres to forecast labels based on maximum a posteriori principle. Then, the second category, referred to as second-order methods, delves into the consideration of pairwise label relationships [19–21]. Nevertheless, it is imperative to acknowledge that label relationships in real-world scenarios tend to be more intricate than simple pairwise associations.

The third category encompasses high-order methods designed to extract more comprehensive insights from the label matrix such as [22–24]. In methods rooted in causal mechanism, such as the M2LC method [25], there is an equitable consideration of features and other labels when labels serve as the focal point. Consequently, discerning causal relationships between target labels and other variables involves the consideration of more than just a pairwise relationship. Owing to the complexity of real-world applications, this often entails a relatively large number of elements. The discourse surrounding high-order relationships predominantly unfold in the context of sparse learning such as [26]. When considering label correlation, it is imperative to account for various scenarios such as semi-supervised learning, partial labels, and missing labels. For instance, Du et al. [27] propose an approach that involves representing instance similarity using weighted graphs constructed from both labeled and unlabeled data. In a similar vein, Hao et al. [28] assume that features with inaccurate labels hold significance and should satisfy certain criteria, such as appearing frequently across multiple groups or exhibiting importance in relation to corresponding labels. Additionally, Zhao et al. [29] decompose the label matrix Y into two components: real labels and error components. They subsequently map the real labels to the feature vector X , while enforcing consistency between features and labels through the introduction of HSIC (Hilbert-Schmidt Independence Criterion) constraints. However, applying these methods directly in multi-view scenarios presents a myriad of challenges, particularly concerning the proliferation of dimensions.

2.2. Multi-view multi-label learning

With the exponential growth of data sources, multi-view learning has garnered significant attention in the research community. Diverging from traditional supervised learning, there have been notable advancements in exploring scenarios where labeled data is limited. For instance, Cao et al. [30] propose the construction of a consensus cluster structure to be learned. They leverage the similarity between the partition matrix derived from the input and the consensus cluster structure to promote stability within the cluster formation. Additionally, they introduce the theory of random walk to capture the multi-order relationships between instances, thereby constraining the adaptive similarity matrix [31]. From a different perspective, achieving a balance between complementarity and consistency is considered a crucial challenge. This can be addressed through techniques such as spectral embedding [32], which facilitates the integration of complementary information from multiple views. Alternatively, minimizing the multiplication of two matrices derived from different views [33] can also enhance the complementarity between views. These approaches provide valuable insights into effectively leveraging multi-view data for better performance. Part of researches in MVML learning primarily address other challenges such as feature weights or class imbalance, thus paying limited attention to label correlation [34–36]. Some works incorporate pairwise label correlation within the learning objective [37]. Another representative works considering high-order label correlation such as [8] and [38] construct a crafted label correlation matrix as vehicles for describing the intricate interconnections among labels. Nevertheless, the matrix transformation remains rooted in the shared aspects of the learning process. In contrast, Zhu et al. [39] adopt a more nuanced method by considering view-specific labels. This mapping of this method is established between latent view-specific labels and the feature matrix in each view. Regrettably, insufficient attention has been given to the varying levels of correlation between labels.

Drawing inspiration from label correlation in multi-label methods, it is plausible to apply this concept to MVML learning by considering the shared portion of the label matrix. The challenge lies in optimizing performance through high-order label correlation while simultaneously considering the view-specific label, particularly the relationship across different layers. Consequently, there remains a substantial scope for exploration within the domain of MVML learning.

3. The proposed method

In this section, the innovative method namely I^2 VSLC is proposed, which takes into account the intra-view and inter-view relationships of view-specific labels. Fig. 1 provides a workflow for the framework of the objective function. Section 3.1 presents the notations employed in I^2 VSLC. Section 3.2 offers an intricate exposition of the derivation process underpinning the proposed methodology. Subsequently, in Section 3.3, we lay down the theoretical underpinnings and apply multiplicative update rules to facilitate the optimization of individual variables. Finally, Section 3.4 delves into a comprehensive analysis of its computational complexity.

3.1. Notations

The basic notations are introduced as follows. Suppose $X = \{x^{(i)}\}_{i=1}^V$ contains V different views, and $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}] \in \mathbb{R}^{n \times d(i)}$, which $x_p^{(i)}$ denotes the p -th instance for i -th view. For the view-specific labels, the notation is expressed as $y^{(i)} = [y_1^{(i)}, y_2^{(i)}, \dots, y_n^{(i)}] \in \{0, 1\}^{n \times c}$. Then, the observed label is set as $Y \in \{0, 1\}^{n \times c}$, and whether the label y_{pj} is annotated depends on the existing of the association with $x_p^{(i)}$.

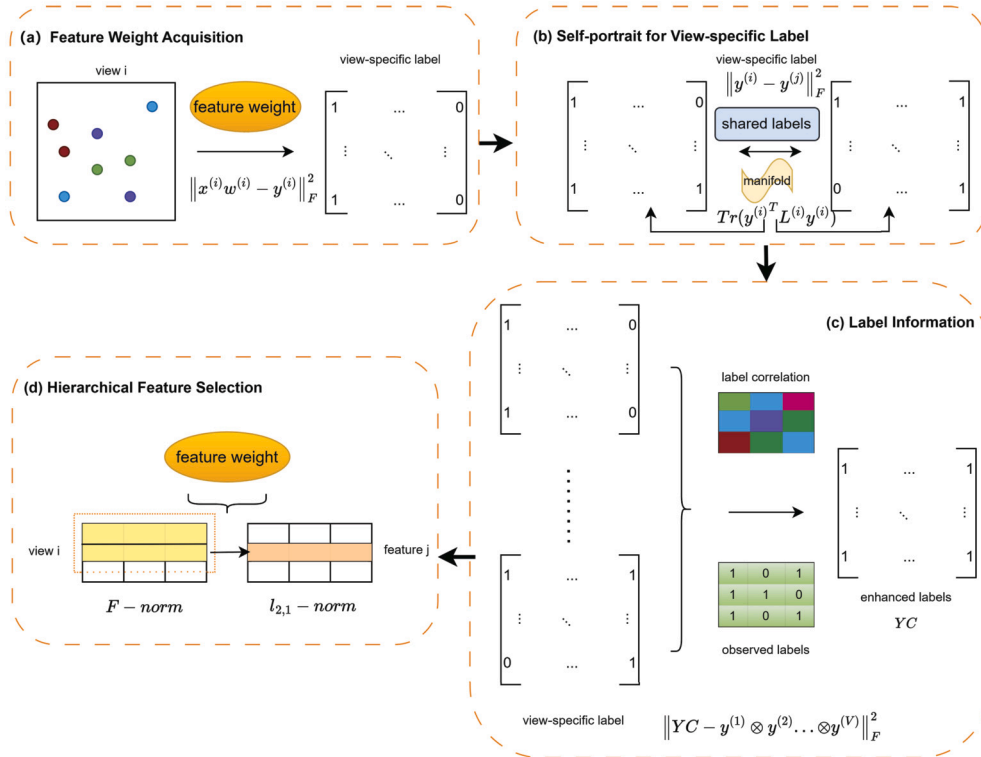


Fig. 1. The framework of the proposed I^2VSLC method revolves around exploring relationships of the view-specific label, which mainly encompasses two main aspects. Firstly, it focuses on the acquisition of view-specific labels via manifold learning, shared label learning, and label correlation techniques. Secondly, it emphasizes the processing of the feature weight matrix.

3.2. Problem formulation

Based on the aforementioned notations, the objective is to construct a unified objective function F_{I^2VSLC} that aims to select informative features based on the relationship among view-specific labels. The proposed formulation encompasses five terms in Formula (1).

$$F_{I^2VSLC} = \mathcal{H}(\cdot) + \mathcal{I}(\cdot) + \mathcal{J}(\cdot) + \mathcal{K}(\cdot) + \mathcal{L}(\cdot). \quad (1)$$

The function $\mathcal{H}(\cdot)$ serves as a mapping function that establishes a connection between features and labels within each view. $\mathcal{I}(\cdot)$ characterizes the intra-view and inter-view relationships, effectively acting as a self-portrait of the view-specific label function. Additionally, $\mathcal{J}(\cdot)$ as a label information function, imposing constraints on the learning direction of the specific label through the utilization of an enhanced label matrix. The functions $\mathcal{K}(\cdot)$ and $\mathcal{L}(\cdot)$ are employed for the weight matrix, serving as regularizers designed to facilitate view-wise feature selection and enable a more refined method to feature selection, respectively.

Initially, one may initiate the derivation by formulating a least squares loss function that effectively captures the relationship between features and view-specific labels:

$$\mathcal{H}(\cdot) = \sum_{i=1}^V \|x^{(i)} w^{(i)} - y^{(i)}\|_F^2, \quad (2)$$

herein, $w^{(i)} \in \mathbb{R}^{d(i) \times c}$ is utilized to denote the feature weights associated with each view. Indeed, the view-specific label represents a novel variable that necessitates learning. Therefore, an examination of its own data structure from an intra-view perspective is undertaken. According to the smoothness assumption [40], we can infer that the similarity of pairwise features has positive correlation with the similarity of pairwise view-specific labels. By leveraging the capabilities of the graph Laplacian, we can embark on the endeavor to grasp the local geometric characteristics inherent in the view-specific label matrix. The elements in affinity matrix $S^{(v)}$ corresponding to v -th view are defined as follows:

$$s_{ij}^{(v)} = \begin{cases} \exp\left(-\frac{\|x_{i.}^{(v)} - x_{j.}^{(v)}\|_2^2}{\sigma^2}\right) & \text{if } x_{i.}^{(v)} \in \mathcal{N}_q(x_{j.}^{(v)}) \text{ or } x_{j.}^{(v)} \in \mathcal{N}_q(x_{i.}^{(v)}), \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

herein, σ represents a predefined parameter, while $\mathcal{N}_q(x_i^{(v)})$ signifies the set of q nearest neighbors of $x_i^{(v)}$ and the similar meaning of $\mathcal{N}_q(x_i^{(v)})$ can be inferred. Accompanied by the affinity matrix as an essential component, the process of learning the local geometric attributes inherent in the view-specific label could be summarized:

$$I(\cdot) = \frac{1}{2} \sum_{i,j=1}^n s_{ij}^{(v)} \|y_i^{(v)} - y_j^{(v)}\|_2^2. \quad (4)$$

According to the equation relationship for $L^{(v)} = A^{(v)} - S^{(v)}$, where $A^{(v)}$ is the diagonal matrix, whose elements are $a_{ii}^{(v)} = \sum_{j=1}^n s_{ij}^{(v)}$. Thus, Formula (4) can be transformed into Formula (5) as:

$$I(\cdot) = \sum_{i=1}^V Tr(y^{(v)T} L^{(v)} y^{(v)}). \quad (5)$$

In essence, there is overlap in the instances described between views, in accordance with the definition of multi-view learning. Consequently, the label matrix within each view exhibits similarities with the label matrices of other views. Basis on this, our objective is to minimize the disparity between any two view-specific label matrices to the greatest extent possible. In order to mitigate any potential impact resulting from the quality of a specific reference point such as average value, a standard point is not employed to measure the difference between two matrices [41]. Then, Formula (6) is defined as follows:

$$\min_{y^{(i)}} \sum_{i=1}^V \sum_{j=i}^V \|y^{(i)} - y^{(j)}\|_F^2. \quad (6)$$

Overall, the self-portrait of the specific label function is defined as:

$$I(\cdot) = \sum_{i=1}^V Tr(y^{(i)T} L^{(i)} y^{(i)}) + \sum_{i=1}^V \sum_{j=i}^V \|y^{(i)} - y^{(j)}\|_F^2. \quad (7)$$

It is important to highlight that the view-specific labels exhibit connections with the observed labels. The characteristics of the view-specific labels are elucidated through considering the relationship between views and the structure for self-view, encompassing the preservation of the shared components while retaining the distinctive data structure complementary to each view. Nonetheless, the consistent relationship between observed labels and view-specific labels still offers the advantageous capability of providing a valid direction. Consequently, the formula can be expressed:

$$\min_{y^{(i)}} \|Y - y^{(1)} \otimes y^{(2)} \dots \otimes y^{(V)}\|_F^2, \quad (8)$$

where the symbol of the OR operation is represented by ' $\otimes$ '. If the label of one instance is annotated in one view, it is expected that the label should be annotated in the same manner across all views. Ideally, the labels in all views would align with the labels present in the observed matrix. However, in reality, the observed label matrix often contains noisy labels due to incorrect or incomplete labeling information. These noisy labels can adversely affect the learning process of view-specific labels and subsequently lead to inaccurate feature selection. Even in the unsupervised scenarios, the issue of noise remains a significant concern. To address this, Xie et al. [42] propose a novel approach involving the construction of a sparse nearest neighbor graph. This graph is utilized to enforce constraints on the coefficient matrix derived from the input data. Subsequently, the denoising coefficient matrix is factorized to obtain the cluster index, which is then used to map the input matrix. This aims to enhance the overall performance by effectively mitigating the impact of noise. Consequently, directly applying it to the regularizer is not advisable. To address this issue, the enhanced label matrix is constructed by utilizing dynamic label correlation. The formula for minimizing the inconsistency between enhanced labels and observed labels is:

$$\min_C \|Y - YC\|_F^2, \quad (9)$$

where $C = [c_{ij}]_{c \times c}$, and the element in the matrix represents the relevance level for i -th label to the j -th label. Thus, the Formula (8) could be transformed as:

$$\min_{y^{(i)}, C} \|YC - y^{(1)} \otimes y^{(2)} \dots \otimes y^{(V)}\|_F^2. \quad (10)$$

In the light of the relationship between enhanced labels and view-specific labels, the function for label information is summarized as follows:

$$\mathcal{J}(\cdot) = \|YC - y^{(1)} \otimes y^{(2)} \dots \otimes y^{(V)}\|_F^2 + \|Y - YC\|_F^2. \quad (11)$$

Finally, the constraint on the feature weights is implemented from two distinct levels, which motivated by [34]. The high-level constraint is guided by the objective of minimizing each element, thereby facilitating the filtration of noisy features within each view and subsequently identifying informative views. To achieve this, the high-level selection leverages the advantages of the Frobenius norm.

$$\mathcal{K}(\cdot) = \sum_{i=1}^V \|w^{(i)}\|_F = \sum_{i=1}^V \sqrt{\sum_{p=1}^{d(i)} \sum_{q=1}^c (w_{pq}^{(i)})^2}. \quad (12)$$

For the low-level feature selection, the $l_{2,1}$ -norm is mainly for measuring the distance among all features, and the informative features will be prominent while other noisy features will be assigned small values [43], where $W = \{w^{(i)}\}_{i=1}^V$ denotes feature weight matrix for all views.

$$\mathcal{L}(\cdot) = \|W\|_{2,1} = \sum_{p=1}^d \sqrt{\sum_{q=1}^c w_{pq}^2}. \quad (13)$$

When the high-level and low-level feature selection techniques are combined, it means that the informative features will be selected from the important views as much as possible.

3.3. Optimization

The unified loss function is consisted of three functions for self-portrait, label information and hierarchical feature selection. The objective function of I^2VSLC is represented:

$$\begin{aligned} F_{I^2VSLC} = & \sum_{i=1}^V \|x^{(i)} w^{(i)} - y^{(i)}\|_F^2 + \alpha \left(\sum_{i=1}^V \text{Tr}(y^{(i)T} L^{(i)} y^{(i)}) \right. \\ & + \sum_{i=1}^V \sum_{j=i}^V \|y^{(i)} - y^{(j)}\|_F^2 \Big) + \beta (\|YC - y^{(1)} \otimes y^{(2)} \dots \otimes y^{(V)}\|_F^2 \\ & + \|Y - YC\|_F^2) + \gamma \sum_{i=1}^V \|w^{(i)}\|_F + \delta \|W\|_{2,1}, \end{aligned} \quad (14)$$

herein, the four trade-off parameters are represented by α , β , γ and δ that ensure the model remains balanced. On one hand, we prioritize the constraint on the learning direction of view-specific labels, incorporating their own attributes and hierarchical relationships. On the other hand, the benefits of hierarchical feature selection are fully leveraged to identify informative features within critical views.

With regard to the logical operation between the enhanced label matrix and the view-specific labels, there exists a possibility to amalgamate the view-specific labels into a unified variable, denoted as Y_{vs} . In light of the inherent characteristic of Y_{vs} , we can represent the equivalent formula as follows:

$$\begin{aligned} y^{(1)} \otimes y^{(2)} \dots \otimes y^{(V)} &= Y_{vs} = \sum_{i=1}^V y^{(i)} \\ &= \begin{pmatrix} y_{vs(11)} & \dots & y_{vs(1c)} \\ \vdots & \ddots & \vdots \\ y_{vs(n1)} & \dots & y_{vs(nc)} \end{pmatrix}. \end{aligned} \quad (15)$$

$$s.t. \quad \forall 1 \leq p \leq n \quad \forall 1 \leq q \leq c \quad y_{vs(pq)} = \begin{cases} 1 & y_{vs(pq)} \geq 1 \\ 0 & y_{vs(pq)} < 1 \end{cases}$$

It is imperative to note that an element within Y_{vs} assumes a value of zero if it signifies that the label lacks annotations from any of the views. Conversely, if it bears a non-zero value, it indicates that the label has been assigned. Thus, the Formula (14) could be transformed as follows:

$$\begin{aligned} F_{I^2VSLC} = & \sum_{i=1}^V \|x^{(i)} w^{(i)} - y^{(i)}\|_F^2 + \alpha \left(\sum_{i=1}^V \text{Tr}(y^{(i)T} L^{(i)} y^{(i)}) \right. \\ & + \sum_{i=1}^V \sum_{j=i}^V \|y^{(i)} - y^{(j)}\|_F^2 \Big) + \beta \left(\left\| YC - \sum_{i=1}^V y^{(i)} \right\|_F^2 \right. \\ & + \|Y - YC\|_F^2 \Big) + \gamma \sum_{i=1}^V \|w^{(i)}\|_F + \delta \|W\|_{2,1}. \end{aligned} \quad (16)$$

Due to the objective function is not smooth, the regularizer such as $\|W\|_{2,1}$ and $\|w^{(i)}\|_F$ needs to be relaxed as $2\text{Tr}(W^T D W)$ and $2\text{Tr}(w^{(i)T} E^{(i)} w^{(i)})$, where both of D and $E^{(i)}$ are the diagonal matrix, $d_{ii} = 1/(2\|W_i\|_2)$ and $e_{jj}^{(i)} = 1/(2\|w_j^{(i)}\|_F)$ [44]. Then, we introduce Lagrangian multipliers to restrict $y^{(i)}$, $w^{(i)}$ and C through the non-negative alternating iterative optimization algorithm, respectively. It is important to emphasize that the regularizer based on Frobenius norm, denoted as $\|B\|_F^2$, can be equivalently expressed as $\text{Tr}(B^T B)$, applicable to the arbitrary matrix B . Therefore, the objective function (16) can be reformulated as follows:

$$\begin{aligned} \min_{y^{(i)}, w^{(i)}, C} & \sum_{i=1}^V \text{Tr} \left[(x^{(i)} w^{(i)} - y^{(i)})^T (x^{(i)} w^{(i)} - y^{(i)}) \right] + \alpha \left(\sum_{i=1}^V \text{Tr}(y^{(i)T} L^{(i)} y^{(i)}) \right. \\ & + \sum_{i=1}^V \sum_{j=i}^V \text{Tr} \left[(y^{(i)} - y^{(j)})^T (y^{(i)} - y^{(j)}) \right] \Big) \end{aligned}$$

Algorithm 1 I^2 VSLC.**Input:** Data matrices $\{X^{(i)}\}_{i=1}^V$; Label matrix Y .**Parameter:** Parameters α, β, γ and δ .**Output:** Set of selected features.

- 1: Initialize $w^{(i)}, C, y^{(i)}$;
- 2: **repeat**
- 3: Update the matrix $w^{(i)}$ in accordance with the formulation specified in Formula (18);
- 4: Update the matrix C in accordance with the formulation specified in Formula (19);
- 5: Update the matrix $y^{(i)}$ in accordance with the formulation specified in Formula (20);
- 6: Update the objective function (17);
- 7: **until** Convergence
- 8: Derive the ordered feature sequence by performing computations on the weights, denoted as $\|W_{(j)}\|_2$ where $j = 1, 2, 3, \dots, d$.

$$\begin{aligned}
& + \beta(Tr[(YC - Y_{vs})^T(YC - Y_{vs})]) \\
& + Tr[(Y - YC)^T(Y - YC)] + \gamma \sum_{i=1}^V 2Tr(w^{(i)T} E^{(i)} w^{(i)}) \\
& + 2\delta Tr(W^T DW) - Tr(\varphi y^{(i)T}) - Tr(\psi w^{(i)T}) - Tr(\mu C^T), \\
& s.t. \{y^{(i)}, w^{(i)}, C\} \geq 0
\end{aligned} \tag{17}$$

where $\varphi \in \mathbb{R}^{n \times c}$, $\psi \in \mathbb{R}^{d \times c}$ and $\mu \in \mathbb{R}^{c \times c}$. Subsequently, we proceed with the incremental update of each variable individually, and a detailed breakdown of this process is elucidated in Algorithm 1.

(1) Fix $y^{(i)}$ and C , update $w^{(i)}$:

The sub-objective for $w^{(i)}$ can be formulated:

$$\begin{aligned}
& \min_{y^{(i)}, w^{(i)}, C} \sum_{i=1}^V Tr[(x^{(i)} w^{(i)} - y^{(i)})^T (x^{(i)} w^{(i)} - y^{(i)})] \\
& + \gamma \sum_{i=1}^V 2Tr(w^{(i)T} E^{(i)} w^{(i)}) + 2\delta Tr(W^T DW) \\
& - Tr(\psi w^{(i)T}).
\end{aligned} \tag{18}$$

Then, the partial derivative for $w^{(i)}$ is shown as:

$$\frac{\partial \Theta}{\partial w^{(i)}} = 2(x^{(i)T} x^{(i)} w^{(i)} + \gamma E^{(i)} w^{(i)} + D^{(i)} w^{(i)} - x^{(i)T} Y) - \psi. \tag{19}$$

And we utilize the KKT conditions into Formula (20), and the results are as below:

$$(x^{(i)T} x^{(i)} w^{(i)} + \gamma E^{(i)} w^{(i)} + D^{(i)} w^{(i)} - x^{(i)T} Y) \circ w^{(i)} = 0, \tag{20}$$

where $\circ$ denotes the Hadamard product. Thus, for $w^{(i)}$, it can be updated by:

$$w^{(i)} \leftarrow w^{(i)} \circ \frac{x^{(i)T} Y}{x^{(i)T} x^{(i)} w^{(i)} + \gamma E^{(i)} w^{(i)} + D^{(i)} w^{(i)}}. \tag{21}$$

(2) Fix $y^{(i)}$ and $w^{(i)}$, update C :

The sub-objective for C can be formulated:

$$\begin{aligned}
& \min_C \beta(Tr[(YC - Y_{vs})^T(YC - Y_{vs})]) \\
& + Tr[(Y - YC)^T(Y - YC)] - Tr(\mu C^T).
\end{aligned} \tag{22}$$

Then, the partial derivative for C is shown as:

$$\frac{\partial \Theta}{\partial C} = 2(Y^T Y + 2Y^T Y C - Y^T Y_{vs}) - \mu. \tag{23}$$

Same as the update principle mentioned above, and the updating rule of C is written as:

$$C \leftarrow C \circ \frac{Y^T Y_{vs}}{Y^T Y + 2Y^T Y C}. \tag{24}$$

(3) Fix $w^{(i)}$ and C , update $y^{(i)}$:

The sub-objective for $y^{(i)}$ can be formulated:

$$\min_{y^{(i)}, w^{(i)}, C} \sum_{i=1}^V Tr[(x^{(i)} w^{(i)} - y^{(i)})^T (x^{(i)} w^{(i)} - y^{(i)})]$$

Table 2

The detailed information for the datasets in our experiments.

Views	3Sources	SCENE	IAPRTC12	Corel5K	ESPgame	OBJECT
View-1(d_1)	BBC(1000)	CH(64)	DH(100)	DH(100)	DH(100)	CH(64)
View-2(d_2)	Reuters(1000)	CM(225)	DHV3H1(300)	DHV3H1(300)	DHV3H1(300)	CM(225)
View-3(d_3)	Guardian(1000)	CORR(144)	GIST(512)	GIST(512)	GIST(512)	CORR(144)
View-4(d_4)	-	EDH(73)	HHV3H1(300)	HHV3H1(300)	HHV3H1(300)	EDH(73)
View-5(d_5)	-	WT(128)	HH(100)	HH(100)	HH(100)	WT(128)
Instances(n)	169	4400	4999	4999	4999	6047
Features(d)	3000	634	1312	1312	1312	634
Labels(c)	6	33	260	260	268	31

$$\begin{aligned}
& + \alpha \left(\sum_{i=1}^V Tr(y^{(i)T} L^{(i)} y^{(i)}) \right. \\
& + \sum_{i=1}^V \sum_{j=1}^V Tr[(y^{(i)} - y^{(j)})^T (y^{(i)} - y^{(j)})] \\
& \left. + \beta Tr[(YC - Y_{vs})^T (YC - Y_{vs})] - Tr(\varphi y^{(i)T}) \right). \quad (25)
\end{aligned}$$

Then, we make the $F = Y_{vs} - y^{(i)}$, $G = y^{(i)} - Y_{vs}$ and V is the number of views as defined in the previous, and the partial derivative for $y^{(i)}$ is shown as:

$$\begin{aligned}
\frac{\partial \Theta}{\partial y^{(i)}} &= 2(1 + \beta)y^{(i)} + \alpha(L^{(i)} y^{(i)} + (V - 1)y^{(i)}) \\
&\quad - x^{(i)}w^{(i)} - \alpha(V - 1)F - \beta(YC + G) - \phi. \quad (26)
\end{aligned}$$

The updating rule of $y^{(i)}$ is written as:

$$y^{(i)} \leftarrow y^{(i)} \circ \frac{\alpha S^{(i)} y^{(i)} + \alpha(V - 1)F + x^{(i)}w^{(i)} + \beta(YC + G)}{\alpha A^{(i)} y^{(i)} + \alpha(V - 1)y^{(i)} + (1 + \beta)y^{(i)}}. \quad (27)$$

3.4. Complexity analysis

The thorough analysis of the computational complexity is conducted, which is associated with the three learning variables in the I^2 VSLC framework, namely $y^{(i)}$, $w^{(i)}$ and C . We make the simplifying assumption that the dimensionality of the feature matrix for each view is denoted by d . The results are as follows:

- (1) Updating $w^{(i)}$ involves four terms of matrix operation, yielding a combined computational complexity of $(O(ndc) + O(d^2n) + O(d^2c))$.
- (2) Updating C involves three terms of matrix operation, and the total complexity is $(O(c^2n) + O(c^3))$.
- (3) Updating $y^{(i)}$ involves eight terms of matrix operation, and the complexity for all terms is $(O(n^2c) + O(nc^2) + O(ndc))$.

Considering the aforementioned complexities, the computational complexity for the training part can be obtained, which is approximated as $(O(d^2n) + O(n^2c))$ per iteration, assuming that $d \gg c$ and $n \gg c$.

4. Experiments

4.1. Experimental setup

4.1.1. Datasets

In the experiment, six datasets are adopted in the benchmark which are popular in the multi-view multi-label scenario. Table 2 illustrates specific meaning of different views, and the quantity of each dimension which represents within the parentheses. The meaning of five views on the two datasets SCENE and OBJECT represents low-level visual features, including histogram, moments, correlogram, edge direction histogram, and wavelet texture, respectively. Five views related to the visual features are extracted from the three datasets Corel5K, IAPRTC12 and ESPgame. The final dataset 3Sources collect three sources to cover distinct new stories from the period February–April 2009: BBC, Reuters and The Guardian. We assign the meaning the number of instances, features, and labels to these variables as n , d and c , respectively, and the range of d is from dozens to several thousands. Additionally, we represent an absent view as ‘-’.

4.1.2. Comparing methods

To valid the effectiveness of I^2 VSLC, we compare it with a selection of prominent MVML learning methods M2LD [7], MSFS [45] and ELSMML [8], as well as multi-label learning methods MoRE [46], MDFS [47], CLML [48] and MRDM [49]. These methods are designed from different entry points such as the relationship between views and the relationship between labels. Meanwhile,

Table 3

Experimental results (mean $\pm$ std) in terms of Average Precision and Coverage, where ‘-’ represents the absent result and the 1st/2nd best results are shown in boldface/underline.

Dataset	I^2VSLC	M2LD	MSFS	ELSMML	MoRE	MDFS	CLML	MRDM
AP $\uparrow$								
3Sources	0.4676 $\pm$ 0.033	0.4368 $\pm$ 0.038	0.4252 $\pm$ 0.035	0.4378 $\pm$ 0.04	-	<u>0.4481 $\pm$ 0.043</u>	0.4232 $\pm$ 0.046	0.4196 $\pm$ 0.041
SCENE	0.7953 $\pm$ 0.008	0.7792 $\pm$ 0.014	0.7887 $\pm$ 0.008	0.7801 $\pm$ 0.01	0.7877 $\pm$ 0.007	0.787 $\pm$ 0.007	0.787 $\pm$ 0.008	<u>0.7929 $\pm$ 0.01</u>
IAPRTC12	0.1436 $\pm$ 0.007	0.137 $\pm$ 0.006	0.1385 $\pm$ 0.005	0.1399 $\pm$ 0.006	0.1263 $\pm$ 0.006	0.1377 $\pm$ 0.006	0.1352 $\pm$ 0.002	<u>0.14 $\pm$ 0.005</u>
Corel5K	0.2361 $\pm$ 0.008	<u>0.2355 $\pm$ 0.008</u>	0.1861 $\pm$ 0.007	0.2317 $\pm$ 0.005	0.1845 $\pm$ 0.023	0.2277 $\pm$ 0.014	0.233 $\pm$ 0.008	0.2339 $\pm$ 0.011
ESPgame	0.2248 $\pm$ 0.005	0.2127 $\pm$ 0.006	0.2113 $\pm$ 0.005	0.2048 $\pm$ 0.005	0.2051 $\pm$ 0.019	0.2123 $\pm$ 0.004	<u>0.2238 $\pm$ 0.004</u>	0.2144 $\pm$ 0.007
OBJECT	0.4873 $\pm$ 0.037	0.4531 $\pm$ 0.026	0.4626 $\pm$ 0.029	0.4632 $\pm$ 0.021	0.4719 $\pm$ 0.031	<u>0.4756 $\pm$ 0.026</u>	0.4748 $\pm$ 0.024	0.4595 $\pm$ 0.02
Coverage $\downarrow$								
3Sources	0.6064 $\pm$ 0.031	0.6397 $\pm$ 0.03	0.6602 $\pm$ 0.037	<u>0.6181 $\pm$ 0.023</u>	-	0.6431 $\pm$ 0.044	0.6727 $\pm$ 0.046	0.6723 $\pm$ 0.042
SCENE	0.4178 $\pm$ 0.015	0.4416 $\pm$ 0.016	0.4271 $\pm$ 0.012	0.4569 $\pm$ 0.016	0.4295 $\pm$ 0.01	0.4331 $\pm$ 0.009	0.4308 $\pm$ 0.012	<u>0.4252 $\pm$ 0.017</u>
IAPRTC12	0.4982 $\pm$ 0.015	0.5131 $\pm$ 0.011	0.5098 $\pm$ 0.009	<u>0.5094 $\pm$ 0.007</u>	0.5877 $\pm$ 0.02	0.5187 $\pm$ 0.021	0.5149 $\pm$ 0.005	0.5097 $\pm$ 0.017
Corel5K	0.4865 $\pm$ 0.015	0.508 $\pm$ 0.008	0.5085 $\pm$ 0.012	<u>0.4898 $\pm$ 0.007</u>	0.5378 $\pm$ 0.012	0.495 $\pm$ 0.016	0.4951 $\pm$ 0.004	0.4918 $\pm$ 0.021
ESPgame	0.5533 $\pm$ 0.021	0.5988 $\pm$ 0.028	<u>0.5838 $\pm$ 0.004</u>	0.6045 $\pm$ 0.031	0.6562 $\pm$ 0.029	0.5876 $\pm$ 0.005	0.588 $\pm$ 0.008	0.592 $\pm$ 0.03
OBJECT	0.2903 $\pm$ 0.022	0.3142 $\pm$ 0.03	0.3005 $\pm$ 0.027	0.3071 $\pm$ 0.017	0.302 $\pm$ 0.03	0.2993 $\pm$ 0.018	<u>0.298 $\pm$ 0.02</u>	0.309 $\pm$ 0.031

they have the common characteristic which is primarily focus on the feature space. The determination of feature weights can be inferred from the objective function, while the performance evaluation can be conducted by varying the number of features based on descending order.

- (1) **M2LD**: the method aims to enhance the discriminative characteristic of the feature space used for model induction by incorporating label information into feature subspace.
- (2) **MSFS**: the method considers the view weight and latent label matrix into the mapping, and performs hierarchical feature selection simultaneously.
- (3) **ELSMML**: the method emphasizes the higher-order relationships among labels, taking into account not only the preservation of low-dimensional structures but also the construction of an enhanced label matrix.
- (4) **MoRE**: the method acts the low-rank projection on two mappings, which are the original data and the Euclidean distance matrices.
- (5) **MDFS**: the method reconstructs the label matrix based on the theory of the similarity local structure. Meanwhile, the new variable preserves the characteristic of the observed labels to minimize the discrepancy.
- (6) **CLML**: the method incorporates both label correlation and instance correlation into the outputs of labels, and considers the common and label-specific features via regularization paradigm.
- (7) **MRDM**: the method constructs a non-linear model through the utilization of manifold regularization and dependence maximization.

4.1.3. Evaluation metrics

To assess the effectiveness of our method, several evaluation metrics are employed, namely Average Precision (AP), Coverage, Hamming Loss (HL), and Ranking Loss (RL). For AP, better performance manifests in the higher value, while the following three metrics act in the opposite direction.

The results of the experiments for the four metrics are obtained, through conducting experiments using five-fold cross-validation. To determine the optimal trade-off parameters, we perform a grid search over the range of $\{10^{-3}, 10^{-2}, \dots, 10^3\}$. The objective of this search is to obtain both the mean accuracy and standard deviation, which serve as indicators for the effectiveness for I^2VSLC .

4.2. Experimental results

4.2.1. Feature selection performance

Table 3-4 presents the performance of the four metrics on the test data. The tables illustrate the results obtained by evaluating these metrics from 1 to 20 percentages of features. The results clearly demonstrate the superior performance for I^2VSLC on the evaluation metrics. Through the above results, it can be seen that:

- (1) The method I^2VSLC on each metric has a bright performance. Statistically analysis reveals that the method MRDM and ELSMML exhibits superior performance, ranking second in 29.2% cases, when comparing with other methods. The reason is that it concentrates on exploring the relationship within labels.
- (2) Our method exhibits a clear advantage over the methods M2LD and MoRE. It is evident that the two methods neglect the relationship among labels. MoRE inclines to establish a mapping between residual structure of features and labels, while M2LD

Table 4

Experimental results (mean $\pm$ std) in terms of Hamming Loss and Ranking Loss, where ‘-’ represents the absent result and the 1st/2nd best results are shown in boldface/underline.

Dataset	I^2VSLC	M2LD	MSFS	ELSMML	MoRE	MDFS	CLML	MRDM
HL $\downarrow$								
3Sources	<u>0.2337$\pm$0.017</u>	0.2491 $\pm$ 0.025	0.2357 $\pm$ 0.017	0.2331$\pm$0.013	-	0.2383 $\pm$ 0.02	0.2471 $\pm$ 0.018	0.2358 $\pm$ 0.017
SCENE	0.0975$\pm$0.004	0.104 $\pm$ 0.006	0.0999 $\pm$ 0.005	0.1068 $\pm$ 0.004	0.1007 $\pm$ 0.005	0.1001 $\pm$ 0.005	0.101 $\pm$ 0.005	<u>0.0979$\pm$0.005</u>
IAPRTC12	0.01537$\pm$0.00	0.01653 $\pm$ 0.00	0.01561 $\pm$ 0.00	0.01551 $\pm$ 0.00	<u>0.0155$\pm$0.00</u>	<u>0.0155$\pm$0.00</u>	0.01554 $\pm$ 0.00	0.0156 $\pm$ 0.00
Corel5K	0.01373$\pm$0.00	0.01465 $\pm$ 0.00	0.01381 $\pm$ 0.00	<u>0.0138$\pm$0.00</u>	0.01381 $\pm$ 0.00	0.01385 $\pm$ 0.00	0.01382 $\pm$ 0.00	<u>0.0138$\pm$0.00</u>
ESPgame	0.0226$\pm$0.00	0.02355 $\pm$ 0.00	0.02362 $\pm$ 0.00	0.02366 $\pm$ 0.00	<u>0.02348$\pm$0.00</u>	0.02365 $\pm$ 0.00	0.02402 $\pm$ 0.00	0.0237 $\pm$ 0.00
OBJECT	0.0556$\pm$0.004	0.0597 $\pm$ 0.003	0.0574 $\pm$ 0.003	0.0597 $\pm$ 0.002	0.0576 $\pm$ 0.003	0.0586 $\pm$ 0.004	<u>0.0573$\pm$0.002</u>	0.0583 $\pm$ 0.003
RL $\downarrow$								
3Sources	0.4593$\pm$0.036	0.5045 $\pm$ 0.039	0.5253 $\pm$ 0.043	<u>0.4875$\pm$0.045</u>	-	0.4943 $\pm$ 0.047	0.5345 $\pm$ 0.055	0.536 $\pm$ 0.051
SCENE	0.0946$\pm$0.004	0.1037 $\pm$ 0.009	0.097 $\pm$ 0.006	0.1041 $\pm$ 0.007	0.0974 $\pm$ 0.005	0.097 $\pm$ 0.006	0.098 $\pm$ 0.006	<u>0.0949$\pm$0.007</u>
IAPRTC12	0.2051$\pm$0.009	0.207 $\pm$ 0.009	0.2093 $\pm$ 0.006	0.2087 $\pm$ 0.006	0.2452 $\pm$ 0.01	0.2094 $\pm$ 0.011	0.2134 $\pm$ 0.003	<u>0.2069$\pm$0.013</u>
Corel5K	0.2306$\pm$0.011	0.2379 $\pm$ 0.005	0.2665 $\pm$ 0.009	<u>0.232$\pm$0.005</u>	0.2613 $\pm$ 0.008	0.2376 $\pm$ 0.012	0.2357 $\pm$ 0.003	0.2343 $\pm$ 0.015
ESPgame	0.2095$\pm$0.005	0.2261 $\pm$ 0.014	<u>0.221$\pm$0.003</u>	0.2315 $\pm$ 0.015	0.2596 $\pm$ 0.015	0.2217 $\pm$ 0.004	0.2286 $\pm$ 0.015	0.2241 $\pm$ 0.017
OBJECT	0.1623$\pm$0.018	0.1791 $\pm$ 0.019	<u>0.1648$\pm$0.017</u>	0.1756 $\pm$ 0.012	0.1661 $\pm$ 0.016	0.1706 $\pm$ 0.015	0.1658 $\pm$ 0.013	0.1744 $\pm$ 0.017

aims to enhance the availability of feature space by sharing a subspace with the label matrix, enabling the learning of a desired label-dependent feature space. Thus, the performance is comparatively inferior when compared to other methods.

- (3) The performance for method MDFS and CLML slightly surpasses that of M2LD and MoRE due to the consideration for the relationship among labels. MDFS incorporates the theory of local similarity structure into the label matrix reconstruction process. Similarly, CLML computes the cosine similarity between labels and utilizes k -nearest neighbors to evaluate the instance correlation for the fitted label matrix. However, these methods take different views as a whole, the relationship between views is ignored. Thus, our method still possesses the advantages.
- (4) Specific to the MVML methods, both MSFS and M2LD take into account the characteristic of each view. Additionally, the part of label correlation and the denoise for label matrix are embedded into the methods. However, these methods primarily focus on exploring commonalities observed between views, resulting in less prominent performance effects.

To provide a clear illustration for the performance of I^2VSLC , we present two datasets showcasing the results for the four metrics in Fig. 2. The curve for AP initially demonstrates an initial upward trend as the number of features increases, eventually reaching a stable level. In contrast, the curves for Coverage and RL exhibit a downward trend initially, followed by a plateau. This observed phenomenon precisely illustrates the efficacy of our feature selection method, highlighting the significant benefits offered by I^2VSLC .

4.2.2. Parameter analysis

The experiment for the sensitivity of parameters $\{\alpha, \beta, \gamma, \delta\}$ is conducted on two datasets in Fig. 3. The X-axis, Y-axis and Z-axis represents the range of parameters, the number of features and the value of the metric, respectively. We employ AP as the evaluation metric for parameter sensitivity on OBJECT and ESPgame. The conclusion from the experimental results can be obtained that ESPgame is relatively stable compared to OBJECT. This is evident from the relatively consistent metric values obtained across different parameter settings, irrespective of the varying number of features. When considering the four parameters, the performance for β and δ are fluctuated rather compared to α and γ . Overall, the performance is less influenced when the parameter value changes.

4.2.3. Convergence

We further verify the convergence of I^2VSLC on three datasets in Fig. 4. From the experimental results, the trend of these curves is consistent, displaying a rapid initial decrease followed by a gradual stabilization above the coordinate axis. These results provide evidence of the convergence and its speed of our method. Furthermore, our method achieves convergence within a few iterations, highlighting its relatively fast convergence speed.

5. Conclusion

In the domain of medicine, there has been a substantial growth in research pertaining to multi views. The term “multiple” primarily refers to diverse data types, such as medical records and images, or distinct images of the same anatomical part. This aligns with the decision-making process of physicians, who rely on multiple sources of information to enhance the diagnostic efficacy of models.

Through experiments conducted on various datasets, the importance of the relationship for view-specific labels can be proved in comparison with other methods. In contrast, I^2VSLC emphasizes the relationships within and between views for view-specific labels. This enables us to explore consistency and diversity across different views concurrently, with our results showcasing the superiority

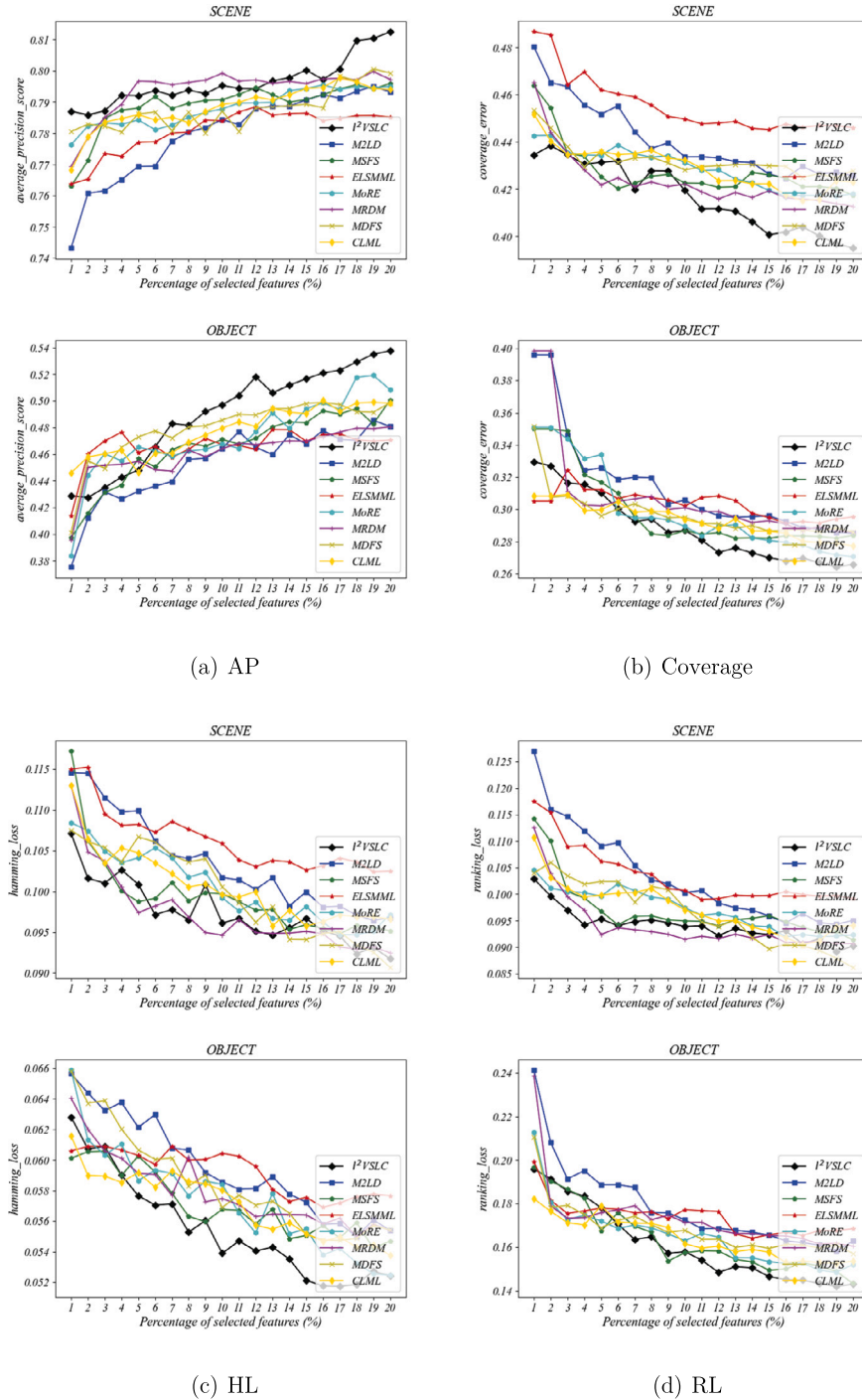
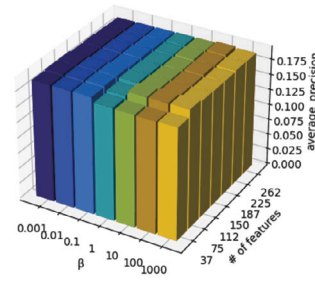
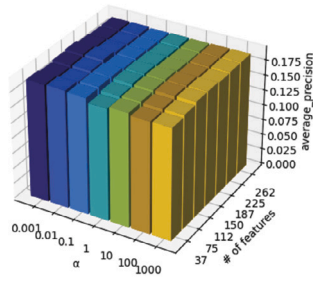
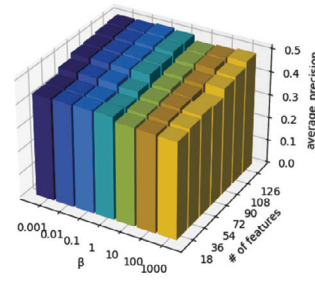
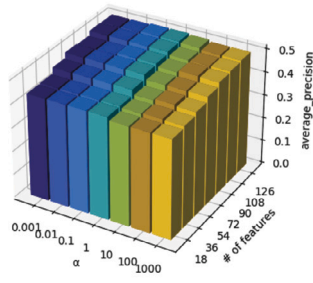
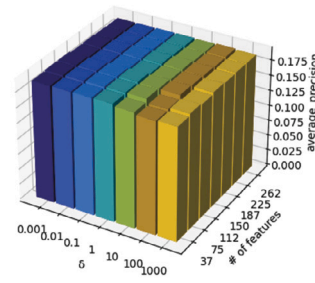
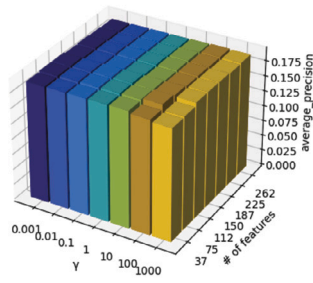
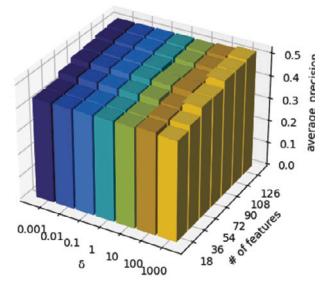
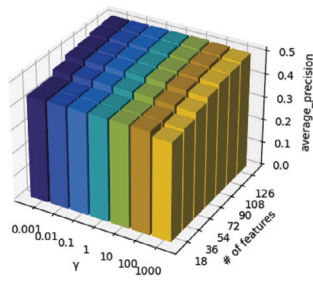


Fig. 2. Seven methods on SCENE and OBJECT in terms of (a) Average precision, (b) Coverage, (c) Hamming Loss and (d) Ranking loss.

of our method over state-of-the-art methods. Inspired by the observation for comparison, our future work aims to further investigate the relationships among labels while minimizing the parameter space requirements. Additionally, we seek to encourage explorations into non-aligned cases.

CRedit authorship contribution statement

Pingting Hao: Writing – original draft, Software, Methodology, Investigation. **Weiping Ding:** Writing – review & editing, Supervision. **Wanfu Gao:** Writing – review & editing, Validation, Conceptualization. **Jialong He:** Writing – review & editing, Supervision.

(a) parameter α (b) parameter β (c) parameter γ (d) parameter δ **Fig. 3.** Parameter sensitivity studies on the OBJECT and ESPgame datasets.

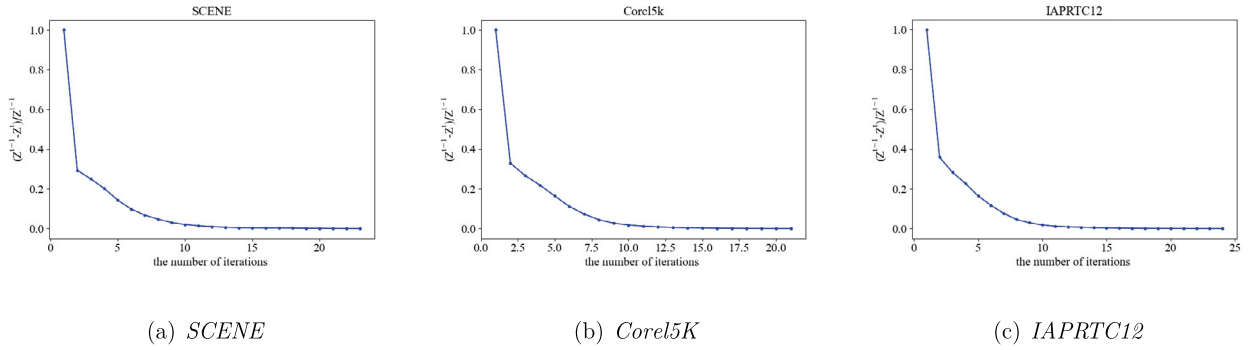


Fig. 4. Convergence curves on SCENE, Corel5K and IAPRTC12.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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